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Data-Driven Improvements in Education

CoachX

The objective of this Project is to evaluate your analytical thinking, problem-solving abilities, and documentation skills using a case study approach. You are required to analyze the provided data, draw insights, and create a detailed report.

Using Data for Improving Educational Outcomes

The Ministry of Education has launched a new initiative to improve student performance in secondary schools across different states. The initiative focuses on enhancing the quality of teaching and learning in subjects like Mathematics, Science, and English. Your task is to analyze the data provided and generate insights that can help in decision-making.

Data Provided:

1. Student performance data across different states.
2. Teacher qualification and training data.
3. School infrastructure and resource availability.
4. Attendance records.
5. Socio-economic background of students.

Tasks

Part 1: Understanding Data:

1. Refer Dataset 1 and identify the issues within the dataset. Document the issues in detail and then move forward to Part 2 of the assignment.

Part 2: Data Analysis:

1. Using the Dataset 2, perform a comprehensive analysis of student performance data across states. Identify trends, patterns, and outliers.
2. Analyze the impact of teacher qualifications and training on student performance.
3. Examine the correlation between school infrastructure and student outcomes.
4. Investigate the relationship between student attendance and performance.
5. Assess how socio-economic factors influence educational outcomes.

Insights Generation:

1. Identify key areas where the initiative has been successful.
2. Highlight regions or states that require additional focus and resources.
3. Provide actionable insights based on the data analysis.

Part 1: Understanding Data

1. Student Table Issues:

- **Data Integrity:** Ensure all student IDs are unique and linked correctly.
- **Date Format Consistency:** Check for uniform date formats.

2. School Infrastructure Table Issues:

- **Data Completeness:** Identify and handle missing values.
- **Data Consistency:** Verify uniform definitions for "internet access" and "library."

3. Socio-Economic Table Issues:

- **Data Completeness:** Ensure all students have recorded socio-economic data.
- **Data Consistency:** Maintain consistent definitions and categories for parent education level and family income.

4. Teacher Data Table Issues:

- **Data Completeness:** Address missing values, especially in the "Training Completed" field.
- **Data Accuracy:** Ensure consistency in recording years of experience and qualifications.

Part 2: Data Analysis

1. Analyze Student Performance Data Across States

Data Preparation:

- **Merge Datasets:** Use Power Query to combine student performance data with socio-economic data using Student ID, and link school and teacher data if needed.
- **Data Cleaning:** Remove duplicates, handle missing values, and ensure date consistency.

Initial Exploration:

- **Pivot Tables and Data Summaries:** Create pivot tables to get an overview of student performance across states.
- **Identify Outliers:** Use visualizations to spot unusual performance scores.

Visual Analysis:

- **Charts:** Create bar charts and line charts to visualize average performance by state.
- **Heatmaps:** Use conditional formatting to identify patterns and disparities.

Statistical Analysis:

- **Descriptive Statistics:** Calculate mean, median, mode, and standard deviation using DAX functions.
- **Correlation Analysis:** Check correlations between variables using Power BI's correlation visual or DAX formulas.

Identify Trends and Patterns:

- **Group Analysis:** Compare performance trends between states with different school infrastructures, teacher qualifications, or socio-economic factors.
- **Temporal Analysis:** Analyze trends over time if data spans multiple periods.

Insights Generation:

- **Highlight Anomalies:** Identify states with performance deviations from the norm.
- **Identify Patterns:** Look for trends like higher performance in states with better infrastructure or higher parental education levels.

Report and Presentation:

- **Create Reports:** Summarize findings in Power BI reports using visualizations and narratives.
- **Dashboards:** Create interactive dashboards for easy data exploration.

2. Impact of Teacher Qualifications and Training on Student Performance

Data Preparation:

- **Merge Datasets:** Combine teacher data (qualifications, training) with student performance data using Teacher ID.

Analysis Steps:

- **Pivot Table:** Create pivot tables to calculate average student performance scores grouped by teacher qualifications.
- **Charts:** Create bar charts to visualize average performance across different teacher qualifications.
- **Statistical Analysis:** Use DAX to calculate correlations between teacher experience, qualifications, training, and performance.

Insights Generation:

- **Identify Trends:** Determine if higher qualifications and training correlate with better performance.
- **Note Correlations:** Identify statistically significant correlations.

Presentation:

- **Create Reports:** Summarize findings with charts and key statistics.
- **Dashboards:** Include interactive elements to explore data.

3. Correlation Between School Infrastructure and Student Outcomes

Data Preparation:

- **Merge Datasets:** Combine school infrastructure data with student performance data using School ID.

Analysis Steps:

- **Pivot Table:** Create pivot tables to calculate average performance scores grouped by infrastructure variables.

- **Correlation Analysis:** Use DAX to calculate correlations between infrastructure variables and performance scores.

Insights Generation:

- **Identify Correlations:** Determine how infrastructure factors like internet access and library availability correlate with performance.
- **Interpret Correlations:** Assess the strength and direction of correlations.

Visualization:

- **Charts:** Create scatter plots and line charts to show correlations.
- **Heatmaps:** Highlight strong correlations.

Presentation:

- **Create Reports:** Summarize findings with visualizations and statistical insights.
- **Dashboards:** Include interactive elements for detailed exploration.

4. Relationship Between Student Attendance and Performance

Data Preparation:

- **Merge Datasets:** Combine attendance data with performance data using Student ID and Date.

Analysis Steps:

- **Calculate Attendance Rate:** Calculate attendance rates for each student.
- **Group Data:** Categorize students based on attendance rates.
- **Pivot Table:** Create pivot tables to calculate average performance scores grouped by attendance categories.

Correlation Analysis:

- **Calculate Correlation:** Use DAX to calculate the correlation between attendance rates and performance scores.
- **Scatter Plot:** Create scatter plots to visualize the relationship.

Insights Generation:

- **Analyze Correlation:** Determine if attendance rates correlate with performance.
- **Identify Trends:** Look for patterns in attendance rates and performance.

Presentation:

- **Create Reports:** Summarize findings with correlation coefficients and key insights.
- **Dashboards:** Include visualizations like scatter plots and bar charts.

5. Influence of Socio-Economic Factors on Educational Outcomes

Data Preparation:

- **Merge Datasets:** Combine socio-economic data with performance data using Student ID.

Analysis Steps:

- **Grouping and Summarizing:** Group students by socio-economic factors.
- **Pivot Table:** Create pivot tables to calculate average performance scores by socio-economic factors.
- **Statistical Analysis:** Calculate descriptive statistics for each socio-economic group.

Correlation Analysis:

- **Calculate Correlation:** Use DAX to calculate correlations between socio-economic factors and performance scores.
- **Charts:** Create visualizations to show relationships.

Insights Generation:

- **Interpret Statistics:** Identify how socio-economic factors influence outcomes.
- **Identify Trends:** Look for patterns like higher performance among students from educated families.

Presentation:

- **Create Reports:** Summarize findings with visualizations and statistical insights.
- **Dashboards:** Include charts and key statistics to support conclusions.

Creating Reports:

- **Comprehensive Analysis Reports:** Use Power BI to create detailed reports covering student performance, teacher impact, infrastructure, attendance, and socio-economic factors.
- **Key Findings:** Highlight trends, patterns, and outliers.
- **Insights and Recommendations:** Provide actionable insights and recommendations based on the analysis.
- **Interactive Dashboards:** Create dashboards for interactive data exploration and presentation

Documentation:

1. Prepare a detailed report summarizing your analysis, insights, and recommendations.
2. Include visual aids (charts, graphs, tables) to support your findings.
3. Ensure the report is well-structured, clear, and concise.

Submission Guidelines

Format: PowerPoint or PDF

Length: 1-20 slides.

Sections: Introduction, Key Findings, Actionable, Methodologies, Approaches, Insights, Conclusions

Tools and Technologies :

POWER BI O/R TABLEAU

Deadline:

Submit your report and presentation within 21 Days from the day you will start.

Notes:

For Project related queries and datasets information is given below .If you find it difficult to understand it you can refer below information to solve this Project or otherwise you can avoid it